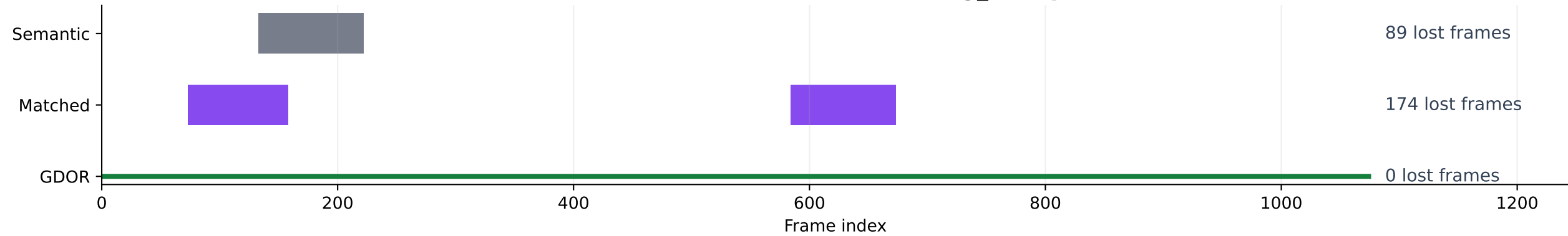
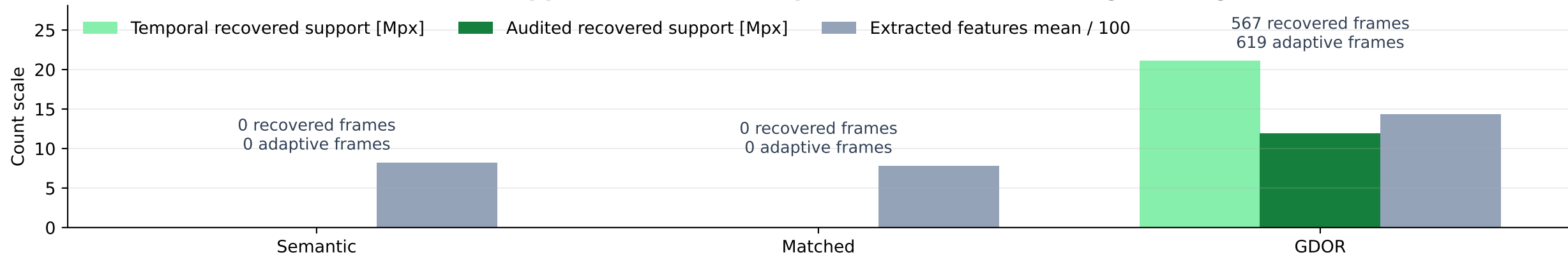


Failure interval, recovered tracking support, and matched control

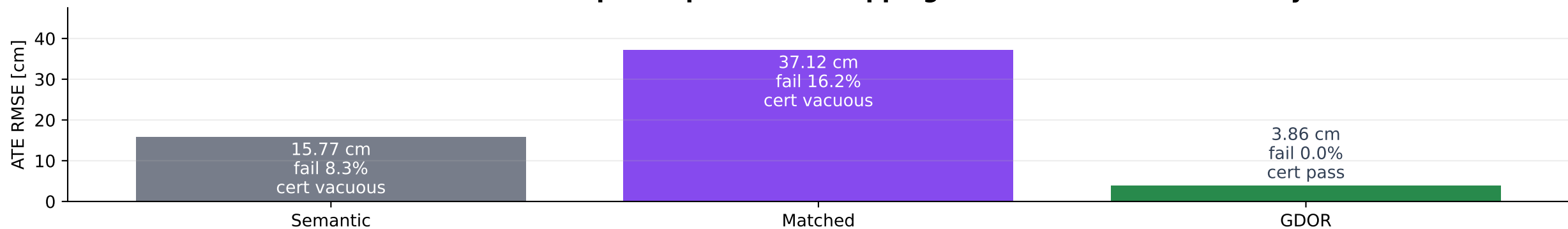
Per-frame failure intervals on TUM sitting_halfsphere, seed 0



Recovered support and feature replenishment are tracking-side signals



Matched control keeps the persistent mapping route but disables recovery



The zero-overlap certificate is a route-invariant diagnostic, not independent ghost-contamination ground truth.